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DEPARTMENT OF DEFENSE

TEST METHOD STANDARD

FOR

ROTORCRAFT STRUCTURAL DEMONSTRATION

AMSC N/A FSC 1520

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FOREWORD

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1. SCOPE

1.1 Scope. This document specifies both the test method requirements for and conduct of the rotorcraft structural demonstration.

1.2 Application. This test method standard applies to all rotorcraft types. For the purpose of this standard, the term *rotorcraft* includes conventional helicopters, tilt rotor, tilt wing, co-axial, and compound helicopters. The terms *rotorcraft* and *aircraft* are used interchangeably herein.

1.3 Objectives. The primary objectives of this rotorcraft structural test method standard are:

a. To establish the essential, tailorable test methods to demonstrate the safe operation of the rotorcraft to the specified maximum attainable operating limits consistent with structural design; i.e., a static test in the air.

b. To provide the necessary test procedures for verifying that loads used in the structural analyses and static tests are not exceeded at the structural design limits of the V-N diagram, or if they are:

c. To provide test and verification methods for establishing the allowable flight envelope and the critical conditions for rotorcraft strength, rigidity, and operation.

NOTE: The structural envelope should meet or exceed the aerodynamic envelope.

1.4 Test activity. The rotorcraft structural demonstration testing may be conducted by either a qualified contractor or a Government entity, in accordance with (IAW) contract requirements. In either case, collaboration is expected between the test activity and the procuring Government agency.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this standard, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

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MIL-STD-3063 *Rotorcraft Structural Integrity Program*

(Copies of this document are available online at https://quicksearch.dla.mil)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

U.S. Army Aviation Engineering Directorate (now Systems Readiness Directorate)

ADS-51-HDBK *Rotorcraft and Aircraft Qualification (RAQ) Handbook*

AMACC *Army Military Airworthiness Certification Criteria*

(Copies of this document are available online at https://avmc.army.mil)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. DEFINITIONS

3.1 Control surface. For the purpose of this standard, includes wings, elevator/stabilator, and vertical fin.

NOTE: Conventional rotorcraft controls (i.e., cyclic, collective, anti-torque) are referred to herein. The same intent would apply to future/alternate control types (e.g., fly-by-wire).

3.2 Conventional rotorcraft. For the purpose of this standard, rotorcraft types having one of the following configurations:

a. Single main rotor with anti-torque tail rotor, and

b. Tandem main rotors.

3.3 Critical Safety Item (CSI).[1] A part, an assembly, installation equipment, launch equipment, recover equipment, or support equipment for an aircraft or aviation weapon system if the part, assembly, or equipment contains a characteristic any failure, malfunction, or absence of which could cause a catastrophic or critical failure resulting in the loss of or serious damage to the aircraft or weapons system, an unacceptable risk of personal injury or loss of life, or an uncommanded engine shutdown that jeopardizes safety.

[1] Definition from Dept. of Army Pamphlet 95-9, “Management of Aviation Critical Safety Items”.

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NOTE: For the purpose of this standard, *aircraft*, *aviation weapon system*, *weapons system*, and similar terms are considered synonymous with *rotorcraft*.

3.4 Limit forward speed (V_H). The maximum level flight speed at structural design gross weight (SDGW) using intermediate rated power (IRP), or as limited by blade stall or compressibility effects.

3.5 Maximum alternate gross weight (MAGW). Defined in the aircraft system performance specification, the procuring Government agency established functional baseline.

3.6 Never exceed speed (V_{NE}). V_{NE} is a "Never Exceed" speed imposed by the procuring Government agency to limit the aircraft envelope until a deficiency is corrected or when a permanent cutoff is indicated.

3.7 Normal load factor (N_Z). Vertical (z-axis) acceleration factor at aircraft CG in aircraft coordinate system.

3.8 Principal structural element (PSE).[2] Structural element that contributes significantly to the carrying of flight, ground, or pressurization loads and the fatigue failure of which could result in catastrophic failure of the rotorcraft.

3.9 Rotor. For the purpose of this standard, includes main rotor(s), tail rotor(s) (anti-torque type and propulsor/forward thrust type), and propellers.

3.10 Safety of flight structure.[3] Structure whose failure could cause loss of the aircraft, or cause severe injury or death, or impair a safety critical function, or cause inadvertent store release. The consequences could occur either immediately upon failure or subsequently if the failure remains undetected.

NOTE: For the purpose of this standard, safety of flight structure includes, but is not necessarily limited to, all CSIs and PSEs.

3.11 Structural design gross weight (SDGW). Defined in the aircraft system performance specification, the procuring Government agency established functional baseline.

3.12 List of acronyms and abbreviations.

ADS	Aeronautical Design Standard
AGL	Above ground level
AvMC	Aviation and Missile Center
CG	Center of gravity
CSI	Critical safety item
DEVCOM	Combat Capabilities Development Command
DLL	Design limit load

[2] Definition from MIL-STD-3063.
[3] Definition from MIL-STD-3063.

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DNE	Do not exceed
GPS	Global positioning system
H_D	Density altitude
H:M:S	Hours:Minutes:Seconds
IAW	In accordance with
IRP	Intermediate rated power
KCAS	Knots calibrated airspeed
KIAS	Knots indicated airspeed
KTAS	Knots true airspeed
MAA	Military airworthiness authority
MAGW	Maximum alternate gross weight
N_G	Gas producer turbine (first stage) speed; a.k.a. N_1
N_P	Power turbine (second stage) speed; a.k.a. N_2
N_R	Main rotor speed
N_Z	Normal load factor
OAT	Outside air temperature
OEM	Original equipment manufacturer
OGE	Out of ground effect
PEO	Program Executive Officer
POC	Point of contact
PSE	Principal structural element
RPM	Revolutions per minute
SDGW	Structural design gross weight
TBE	To be established
V_{DL}	Design dive limit speed
V_H	Maximum attainable level flight speed at intermediate rated power
V_{NE}	“Never Exceed” speed

4. GENERAL REQUIREMENTS

4.1 General. Conduct a structural demonstration with structural instrumentation necessary to substantiate the airworthiness of the rotorcraft and to provide a formal demonstration of compliance with the structural requirements of the rotorcraft system specification. The general requirements for the structural demonstration, as specified herein, may be modified as necessary for specific aircraft models and the applicable development lifecycle phase. Refer to 6.3 for tailoring guidance. Carefully consider each of the many parameters involved in the design of a successful structural demonstration.

4.2 Aircraft configuration. On structural demonstration aircraft, the structure and all systems which could directly or indirectly affect the results obtained during the test should be closely representative of the aircraft product baseline configuration. The use of dummy equipment, having the correct shape (i.e., volume), mass, and inertial properties to simulate externally mounted equipment if the production hardware is not available, may be permitted. Approval by the procuring Government agency is required before the addition of necessary ballast to attain specified CG locations and the installation of special test instrumentation. In particular, once the demonstration commences, no changes should be made to any system unless potential or actual

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adverse aircraft response or malfunction is evident from the testing conducted up to that point, and is of a type not previously observed in prior testing. No special installation or deletion of any kind should be made without Government approval. If necessary, a decision to continue or discontinue the demonstration shall be made by the Government based on the nature and severity of the change.

4.3 Supporting data. Obtain the data discussed in 4.3.1 and 4.3.2 required to support the rotorcraft structural demonstration in sufficient time to allow adequate review and resolution of potential or actual problem areas prior to initiation of testing.

4.3.1 Analyses and ground tests. Conduct the following rotorcraft analyses and ground tests. Record the resulting data IAW MIL-STD-3063 and the Structures section of AMACC prior to initiating the rotorcraft structural demonstration testing:

a. Stress and Fatigue Analyses *(See 5.2.3.3 Finite Element Models, AMACC p. 68. Also see 7-6.1.2 Fatigue Integrity, 7-6.1.3 Structural Integrity Verification, and 7-6.3 Fatigue Loading in ADS-51-HDBK, Rotorcraft and Aircraft Qualifications, p. 7-7 – 7-13.)*

b. External Loads (all flight, ground, and weapons delivery conditions) *(See 7-6.2 STATIC LOADING, p. 7-9 – 7-10, 7-6.3 FATIGUE LOADING, p. 7-10 – 7-12, ADS-51-HDBK, Rotorcraft & Aircraft Qualifications. Also see 5.2.2 Loads analysis, MIL-STD-3063, p. 27-28.)*

c. Structural Description and Design Criteria *(See 5.1.9 for Structural description; and 5.1.3 Structural design criteria, MIL-STD-3063 Rotorcraft Structural Integrity Program Standard Practice, p. 17-26)*

d. Strength Summary and Operating Restrictions *(See F-2.1.5.4 Strength Summary; and F-2.1.7 Operating Restrictions ADS-51-HDBK, Rotorcraft and Aircraft Qualifications, p. F-5 and F-7, respectively. Also see 5.4.2 Strength summary and operating restrictions, MIL-STD-3063, p. 48)*

e. Mechanical Instability and Flutter Analysis *(See 4-9.5 AERODYNAMIC PROPERTIES, Subparagraph 3. Flutter Analysis reports, ADS-51-HDBK, Rotorcraft and Aircraft Qualifications, p. 4-27 – 4-28. Also see 9-5 DYNAMIC STABILITY, p. 9-27 – 9-29 and 9-7.1 GROUND VIBRATION TESTS, p. 9-40. Also see 5.2.13 Aeroelastic and aeroservoelastic analysis, MIL-STD-3063, p. 32.)*

f. Blade Stall and Compressibility Analysis *(See 5.2.13.1 Blade stall and compressibility analysis, MIL-STD-3063, p. 32.)*

g. Airframe and Dynamic Component Static Tests *(See 9-2.1 STATIC TEST PROGRAM, ADS-51-HDBK, Rotorcraft and Aircraft Qualifications, p. 9-3 – 9-4. Also see 5.3.1 Static tests, MIL-STD-3063, p. 36.)*

h. Landing Gear Drop Tests *(See 8-6.1 Drop testing, ADS-51-HDBK, Rotorcraft and Aircraft Qualifications, p. 8-15 – 8-16. Also see 5.3.8 Drop tests, MIL-STD-3063, p. 42.)*

i. Airframe and Dynamic Component Fatigue Tests *(See 7-6.1.2 Fatigue Integrity, and 7-6.3 Fatigue Loading in ADS-51-HDBK, Rotorcraft and Aircraft Qualifications, p. 7-7 – 7-13. Also see 5.3.3 Safe life tests, 5.3.4 Enhanced safe life tests, 5.3.5 Design service life tests, MIL-STD-3063, p. 38-39.)*

j. Structural Demonstration Test Plan, including Do Not Exceed (DNE) limits *(See 6.8. Also see MIL-STD-3063, Rotorcraft Structural Integrity Program Standard Practice 5.3.10.2 and 5.3.14, p. 45 – 47.)*

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k. Structural Demonstration Maneuver Standards Guide *(See 5.3.10 Structural flight tests, p. 45 and 5.3.10.2 Structural demonstration, pp.45-46, MIL-STD-3063 Rotorcraft Structural Integrity Program Standard Practice.)*

4.3.2 Flight tests. Complete the Flight Loads Survey IAW MIL-STD-3063 Flight loads survey section and AMACC Flight Loads Survey section prior to the structural demonstration.

4.4 Rate and sequence of control inputs. Control input rate is proportional to load magnitude. The sequencing of control inputs can further affect loads. For these reasons, perform the maneuvers by specified movement of the controls. Hold the controls position(s) the length of time specified to attain the maximum or specified flight parameter(s) and return them to the position specified for level coordinated flight, in not greater than the specified time. Specific input details are required and are necessary to gain consistency in the results by eliminating, to every practical degree, differences in pilot technique and expertise. Document these input details in the Structural Demonstration test plan (see 4.3.1 j.), and fully define them in the Structural Demonstration Maneuver Standards Guide (see 4.3.1 k.).

4.5 Gross weight and CG location(s). Complete the demonstration at test conditions representing a variety of gross weights and CG locations as required by the procuring Government agency. The combination of conditions should represent structurally significant design points meant to define the design envelope. Demonstrations are predominantly performed in primary mission configuration at a typical mission weight, such as the structural design gross weight (SDGW). Additional test points should also be conducted in alternate configurations, such as the maximum alternate gross weight (MAGW). Perform testing in both a clean and stores configuration when the aircraft is designed with external stores. The CG should be at the most critical location(s) for each gross weight being investigated.

4.6 Atmospheric conditions. Conduct all flights in conditions specified for the maneuver of the aircraft under test, its specified operational environments, and the procuring Government agency.

4.7 Trim and altitude conditions. For the test maneuvers,

a. Trim the aircraft in steady, unaccelerated flight at the required test airspeed ±5 knots.

b. The test altitude should be the minimum safe (or higher) altitude for the test condition. This is a static test in the air with the expressed purpose of loading the entire aircraft, not just the rotor systems. The test conditions should include altitudes up to the "all engines operating" service ceiling as specified by the procuring Government agency.

4.8 Instrumentation and data analysis. Obtain evidence of compliance with the test requirements (4.3.1 j.) or prior attainment of any of the critical limits of 5.6 by assessing time history data delineating the parameters pertinent to each demonstration maneuver (4.3.1 k.). Use telemetering for critical parameters that require instantaneous load information, thereby increasing flight safety and expediting test progress. Refer to AMACC Instrumentation and Calibration for Structural Flight Testing section for guidance. Prior to procurement and installation of instrumentation, consider the following:

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a. Collaboration with appropriate procuring Government agency to ensure compatibility of proposed instrumentation and telemetering equipment to meet data deliverable requirements. If possible, modify existing equipment to be compatible, and purchase new systems to be compatible.

b. Incorporate backup power to assure continuity of transmission of telemetered data in the event of a power failure.

c. Redundant source of data for critical parameters, where practical, to prevent loss of a data flight due to instrumentation malfunction.

d. The extent to employ telemetering during the test.

e. Protection and recovery of the onboard instrumentation records in the event of aircraft loss.

4.9 Instrumentation package demonstration. Conduct an instrumentation package demonstration to establish the adequacy of the instrumentation package performance and operating specifications to document fulfillment of the structural demonstration requirements. The instrumentation package should include the capability to measure and record all parameters necessary to document compliance with the demonstration requirements and to substantiate the structural integrity of the aircraft. Tailor the complete instrumentation package to fit the specific aircraft. As a minimum, and as applicable, instrumentation should record:

a. Pilot acceleration (vertical and lateral)

b. Control positions (longitudinal, lateral, directional, collective, etc.)

c. Attitudes (roll, pitch, and yaw)

d. Rates (roll, pitch, and yaw)

e. Accelerations at CG (vertical, lateral, and longitudinal)

f. Rotor pitch angles; rotor flapping; rotor lead-lag

g. Speeds. Indicated (KIAS), calibrated (KCAS), and true (KTAS) airspeeds, measured in relatively undisturbed air, such as with an air data boom. Ground speed, measured by pace vehicle or GPS, as applicable for required accuracy.

h. Altitude (H_D and AGL)

i. Engine(s) parameters (N_P, N_G, N_R, Torque)

j. Clock (H:M:S), event marker, and run number

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k. Rate of climb/descent

l. Fuel quantity

m. Rotor blade bending (flapwise and chordwise at critical locations), torsion, and measured axial load (not derived).

n. Actuator loads and positions for all control mechanisms

o. Rotor pitch link, drive arms, drag links, and lift link loads

p. Rotor mast torque

q. Rotor mast bending

r. Rotor azimuth and RPM

s. Airframe loads or stresses (critical beam caps, shear webs, struts, support fittings, engine mounts, tail boom, control tubes, transmission mounts, rotor mounts, etc.)

t. Landing gear loads or stresses

u. Control surface loads or stresses

v. Miscellaneous - vertical sink speed, angle of attack, angle of sideslip, OAT

w. Any parameter necessary to calculate a derived parameter

4.10 In-flight visual monitoring. The structural demonstration should include cockpit audio and video monitoring and recording, as well as video data coverage of various critical locations of the aircraft during compliance tests. Use a chase camera and field camera to record the test aircraft throughout execution of the structural demonstration.

4.11 Ground-based real-time telemetry monitoring. Prepare a table of critical components and their maximum allowable peak and oscillatory parameters (e.g., loads, stresses, or strains). Each should be monitored during all test phases and should not be exceeded during the demonstration. Inadvertent exceedance requires returning the aircraft to the flight line for all necessary inspections before continued flight is permitted.

5. DETAILED REQUIREMENTS

5.1 General. The purpose of this section is to specify the maneuvers, flight techniques, and detailed compliance criteria necessary to satisfy the rotorcraft structural demonstration requirements.

5.2 Tailoring maneuvers. Historically, flight demonstrations for conventional rotorcraft have included, but were not limited to, symmetric pullups, pushovers, rolling pullups, dynamic yaws,

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sideslips, autorotation, slope landings, hard landings, nap of the earth, and any specific combat maneuvers. Tailor the details of each test condition to the specified rotorcraft type defined in the structural portion of an integrated test plan. See 5.3 for selected detailed test maneuvers. Where appropriate, use described, alternate maneuver techniques that best satisfy the intent of the demonstration. The Government may permit minor variations from the described techniques, which best serve the purpose of the rotorcraft structural demonstration. Refer to 6.3 for tailoring guidance. Thoroughly substantiate the approved variations from the following described and specified alternate techniques.

5.3 Rotorcraft test maneuvers. Each test program should establish the maneuvers that evaluate the maximum design limit loads for critical aircraft components within the structural design envelope. A sample matrix of test conditions and configurations for various maneuvers is presented in TABLE I. Application of collective control during load factor generating maneuvers should be used to take maximum advantage of rotor speed build or decay. Phasing of control inputs should normally be determined during the structural demonstration buildup flights. Rotor speed should be controlled to preclude excursion beyond limits already established for the demonstration. Avoid N_R exceedances which disallow the test data, or that require returning the aircraft to the flight line for inspection.

5.3.1 Airspeed/rotor speed. Establish the maximum allowable air speed as a trim point at the maximum and minimum allowable steady state rotor speed. Conduct this maneuver at the allowable steady state power extremes (maximum power-on and power-off) and at least one intermediate power point. Fly all subsequent maneuvers at 100 percent[4] rotor speed unless the maximum and minimum allowable rotor speed is greater than 3 percent of the nominal. For this case, demonstrate the extremes for all "g sensitive" conditions except the Autorotative Overspeed Pullout.

5.3.2 Symmetrical pullout. Establish level flight at the target airspeed and maintain that collective setting. Using cyclic, decelerate to below the target airspeed; then apply forward cyclic to accelerate to the target airspeed in a nose down attitude. At the target airspeed, apply phased aft cyclic and up collective until the maximum attainable load factor is obtained. Terminate the maneuver when a limiting parameter is met. Refer to TABLE I for sample target airspeeds.

5.3.2.1 Alternate techniques.

a. When entry conditions require near maximum collective settings using the above technique, reduce collective control to the collective position required for maximum endurance air speed. Up to contingency power may be used during maneuver execution.

b. Using the normal technique, use an oblique entry when pitch attitude, pitch rate, or both will obviously restrict attaining the maximum possible load factor. Once the entry condition is established as above, the aircraft is rolled to a predetermined attitude. After roll attitude is established, longitudinal and collective controls are applied as if in a level pullup. This is a difficult maneuver requiring close test team coordination and good timing

[4] Or other nominal operating RPM, as applicable.

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for proper execution. Do not use technique 5.3.3 to satisfy the symmetrical pullout requirement.

5.3.3 Rolling pullout. Establish level flight at the target airspeed and maintain the collective setting. Using cyclic, decelerate to below the target airspeed; then apply forward and lateral cyclic to accelerate to and arrive at the target air speed in a nose down, banked attitude. At the target airspeed, make phased longitudinal, lateral, directional, and collective control inputs to achieve a coordinated rolling pullout at the specified load factor. Maximum roll rate and peak load factor should occur as the aircraft passes through the wings level attitude. Recover once peak load factor is achieved. Conduct rolling pullouts to the left and to the right. Refer to TABLE I for sample target airspeeds.

5.3.4 Symmetrical pushover. Establish level flight at the target airspeed. With collective control fixed at the level flight setting, accelerate to above target airspeed. Then pitch aircraft nose up in a decelerating attitude. As airspeed decreases, displace flight controls to achieve the minimum normal load factor with aircraft in level flight attitude at the target airspeed. Large changes in aircraft attitude during this maneuver are not normally required. Refer to TABLE I for sample target airspeeds.

5.3.4.1 Alternate techniques. Explore two alternate techniques during the buildup flights to determine if there are limitations: collective-only and cyclic-only "low g" maneuvers.

a. Conduct collective-only symmetrical pushover from level flight at the target airspeed. Apply collective control to achieve minimum load factor while adjusting the other flight controls to maintain aircraft pitch, roll, and yaw attitudes. Do not permit rotor speed to exceed pre-established limits.

b. Conduct the cyclic-only symmetrical pushover in the same manner as the basic maneuver except collective control should remain fixed (small collective adjustment may be necessary to keep rotor speed within limits).

5.3.5 Autorotative pullout. Enter this high airspeed maneuver from an autorotative dive at the design maximum power-off RPM or full down collective. Initiate a cyclic flare as the target airspeed is approached, and input sufficient collective to achieve the transient power-off rotor RPM. The limit normal load factor and target airspeed should be attained when the aircraft reaches a level flight attitude. The airspeed maneuvers below TBE knots[5] are entered from stabilized autorotative descents at design maximum power-off rotor RPM or that RPM attainable with minimum collective. The maneuver consists of a rapid movement of the collective to the upper limit of its travel, a pause sufficient to allow the normal acceleration to peak, followed by a recovery to powered flight. The cyclic and directional controls are applied as required to maintain coordinated, steady heading flight. Refer to TABLE I for sample target airspeeds.

[5] See FIGURE 1. Typically, the lowest speed of Zone B in a height-velocity curve.

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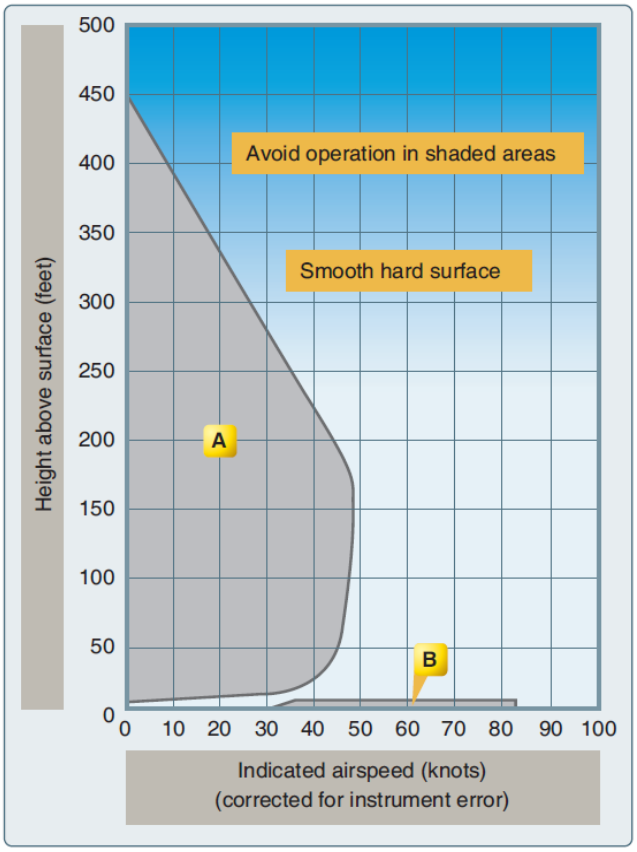


FIGURE 1. Typical height-velocity diagram.[6]

5.3.6 Jump takeoff. The jump takeoff is made from a zero ground speed, "light on wheels," maximum operational rotor RPM condition in winds not greater than 5 knots. Accomplish this maneuver by raising the collective rapidly to the maximum usable, pausing briefly at that position a sufficient time to record the peak normal acceleration before recovering and without the rotor RPM drooping below the transient power-on limit. Apply cyclic and directional controls as required to maintain a non-turning, level attitude. Also demonstrate a jump takeoff from a minimum collective condition. The method which produces the greater normal load factor should be the demonstration end point.

5.3.7 Steady sideslip. Yaw the rotorcraft to the specific sideslip angle using directional control (up to full displacement) and opposite lateral control to maintain steady, non-turning flight. Perform this maneuver with the collective control fixed at the setting for steady trimmed level flight (or steady decent if above V_H) at the target airspeed. Descent is permissible to maintain target airspeed. Execute the steady sideslip in both left and right yaw. Refer to TABLE I for sample target airspeeds.

[6] Credit: U.S. Dept. of Transportation, Federal Aviation Administration, 2012, FAA-H-8083-21A, *Helicopter Flying Handbook*, Chapter 11, Figure 11-3. Retrieved 26 August 2019 from https://www.faa.gov/regulations_policies/handbooks_manuals/aviation/helicopter_flying_handbook/.

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5.3.8 Dynamic yaw. Initiate dynamic yaw from steady level flight (or steady descent at speeds above V_H) with collective fixed and at design maximum power-on RPM. Rapidly yaw the aircraft in one direction and apply opposite pedal control allowing the aircraft to reach the transient sideslip limit. Use full control displacement to accomplish the reversal but hold only until attaining the maximum angular yawing velocity, or until recovery is required to preclude exceeding the transient sideslip limits. Perform this maneuver to the left and right. Refer to TABLE I for sample target airspeeds.

5.3.9 Turn reversals. Enter the maneuver with the collective set at the trimmed level flight setting for the target airspeed. Bank the aircraft in a steep turn so that, after applying full lateral control to roll the aircraft level, it achieves maximum roll rate at the approximate time the aircraft reaches a level flight attitude. Apply longitudinal, cyclic, and directional control to maintain coordinated flight. Control inputs to the full stop positions may be necessary. Demonstrate this maneuver in both directions. Refer to TABLE I for sample target airspeeds.

5.3.10 Quick stop. From a stabilized hover, initiate forward flight at IRP. Apply a combination of cyclic, collective, and pedal controls to return to a stabilized hover in minimum time. The condition is met when rotor speed cannot be maintained below maximum transient power-on or maximum autorotative RPM, control stop contact, or a structural limit is reached. Maintain altitude[7] and heading during the maneuver from hover to hover.

5.3.11 Hard landings (vertical and run-on). Demonstrated landings should be from a hover, on a specified side slope, and in a run-on to the design limit sink speed at ground contact. Forward speed should include the maximum allowable for landing gear design. The instrumentation should document the rate of descent.

5.3.12 Autorotation overspeed. Stabilize the aircraft in an autorotation at the target airspeed at 100 percent RPM. Apply aft cyclic and accelerate the rotor to TBE percent[8] of design limit power-off RPM while increasing load factor to the maximum attainable. Collective may be used during the maneuver to increase load factor and control RPM. The goal is to achieve maximum attainable load factor at TBE percent[8] RPM. Do not exceed maximum N_Z. Repeat the maneuver at the target airspeed, but the load factor should be 1.0. All drive system components should have critical rotational frequencies greater than TBE percent[8] RPM equivalent on the rotor.

5.3.13 Hover turn arrestment. From a stabilized hover turn at maximum design turn rate, apply full pedal to stop the turn in the shortest possible time. Maneuver is satisfied when the directional force, pedal stop time, or turn arrestment time limit is reached. Values should be defined when the performance/operational capability of the rotorcraft is understood.

5.3.14 Altitude maneuvers. Perform selected maneuvers at high altitude for the purpose of establishing any structural limitations and the corresponding V-N diagram. This altitude should be defined when the performance/operational capability of the rotorcraft is understood. TABLE II presents a sample matrix of these high-altitude maneuvers. Demonstrated load factors should not be adjusted for density altitude and gross weight at the conclusion of the test.

[7] Sufficient altitude to ensure no ground contact is made by tail boom or aft end of aircraft during the maneuver.
[8] Greater than 100 percent, specified in the structural design criteria.

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TABLE I. Sample structural demonstration maneuvers.[1]

MANEUVER	GROSS WEIGHT[2]	TARGET AIRSPEED	TARGET
Maximum Airspeed	SDGW, MAGW	V_{DL}[3]	Max/Min N_R
Symmetrical Pullout	SDGW SDGW, MAGW SDGW, MAGW SDGW, MAGW	$0.4V_H$ $0.7V_H$ $1.0V_H$ V_{DL}	V-N diagram
Rolling Pullout Left and Right	SDGW SDGW, MAGW	$0.6V_H$ $1.0V_H$	V-N diagram
Pushover	SDGW, MAGW SDGW	$0.8V_H$ V_{maxRC}[7]	V-N diagram
Autorotative Pullout	SDGW	$0.2V_{MA}$[4] $0.6V_{MA}$ $1.0V_{MA}$	V-N diagram
Jump Takeoff	SDGW	0	V-N diagram
Steady Sideslip Left and Right	SDGW SDGW	$1.0V_H$ V_{DL}	Max Yaw Angle
Dynamic Yaw Left and Right	SDGW SDGW SDGW	$0.5V_H$ $0.8V_H$ $1.0V_H$	Transient Sideslip Limit
Turn Reversals Left and Right	SDGW SDGW SDGW	$0.6V_H$ $0.8V_H$ $1.0V_H$	Max Roll Rate and Max Bank Angle
Quick stop	SDGW	40 knots	
Hard Landing	SDGW	Hover Max run-on	V_v Limit Sink Speed[5] V_V and V_G at Ground Contact[6]
Autorotative Overspeed	SDGW	TBD	115%[8] N_R Max N_Z
Hover Turn Arrestment	SDGW	Hover OGE	Shortest Time

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[1] These maneuvers should be accomplished at 3,000 feet ±1,000 feet density altitude except jump takeoff and quick stop which will be accomplished at prevailing density altitude.

[2] Center of Gravity (CG) should be the critical CG(s) determined from the flight load survey.

[3] V_{DL} = 1.2 V_H (knots).

[4] 0.2V_{MA} or 35 knots, whichever is greater. V_{MA} is maximum speed for stabilized autorotation.

[5] V_V is the limit vertical sink speed.

[6] V_G is the limit ground speed

[7] V_{maxRC} is the airspeed for maximum rate of climb.

[8] Unless otherwise specified in the structural design criteria.

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TABLE II. Sample high altitude maneuvers.

MANEUVER	WEIGHT	AIRSPEED	TARGET
Maximum Airspeed	SDGW	V_{DL}	Max N_R Min N_R
Symmetrical Pullout	SDGW	V_H V_{DL}	Max N_Z
Rolling Pullout Left and Right	SDGW	V_H	Max N_Z
Pushover	SDGW	V_{maxRC}	Max N_Z
Autorotative Pullout	SDGW	V_{MA}	Max N_Z

5.4 Future rotorcraft configuration test maneuvers. In addition to those listed in 5.3, additional maneuvers may be necessary to fully evaluate future rotorcraft design configurations. Test maneuvers for future rotorcraft configurations should be developed that evaluate the maximum design limit loads in critical aircraft components.

5.5 Maneuver requirements. The following are required when performing demonstration maneuvers.

5.5.1 Control input time. For fly-by-wire or computerized flight control systems, the time required for control movement should be the maximum input rate allowed by the flight control system. For mechanical linkage systems, the time required for control movement should be the fastest input rate by the pilot.

5.5.2 Rotor RPM. Rotor RPM should be the design maximum power-on or power-off, as applicable, unless the contractor has demonstrated that an alternate rotor speed is more critical. Include the maximum/minimum transient power-on/power-off RPM in the structural demonstration test plan.

5.6 Test cutoff limits. Compliance with the requirement to demonstrate the critical limits is accomplished when any one, or combination, of the test cutoff limits specified in 5.6 have been attained and approved by the procuring Government agency. Nothing in this standard is intended to limit the authority of the test director or pilot from aborting a flight or terminating a test condition for any cause whenever deemed prudent in their judgment, singly or jointly. Evaluate potential limit exceedances during post-test review and consider repeating maneuver(s) on alternate day or with alternate technique to demonstrate greater performance or load factor. If limits are exceeded, test results may no longer be valid. Document limits for inclusion in the -10 operator's manual.

5.6.1 Limit load factor. The limit load factor is obtained from the structural design envelope (V-N diagram). An aerodynamic limit may be encountered during any particular maneuver and is a valid

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cutoff. However, no maneuver should be artificially terminated based on the aerodynamic limits used for design. One fundamental purpose of the demonstration is to establish where the aerodynamic boundaries lie and if they are indeed limiting. Demonstrated load factors should not be adjusted for density altitude and gross weight at the conclusion of the test.

5.6.2 Rotor blade stall. Rotor blade stall is evidenced by a decreasing load factor with increasing collective, aft cyclic, or both or by stall-induced loads approaching component allowable loads. Identify and document aircraft configuration and flight parameters that produce an unacceptable stall condition. IAW MIL-STD-3063 Dynamics criteria section, "…avoid introducing deleterious oscillatory loads due to retreating blade stall or advancing blade compressibility in sustained steady-state maneuvers such as level flight or other steady-state conditions, such as those producing a load factor less than 1.25g."

5.6.3 Excessive angular rates, accelerations, and attitudes. Establish the OEM's angular rates, acceleration, and attitude limits of safe operation for the rotorcraft. Develop maximum rates, accelerations, and attitudes considering combinations and attitudes based on values determined to be at or approaching the limit of safe recovery. Consider vibration levels, controllability, potential overshoot, etc., in determining the rotorcraft Do Not Exceed (DNE) limits.

5.6.4 Excessive vibratory levels. Identify and document vibration identified by the flight crew or instrumentation that is determined to be excessive by the flight crew or test director.

5.6.5 Specified control rates or displacements. Identify and document limits of full travel control displacement (control stop contact) achievement within the required time, and whether an excessive increase in pilot effort is required to move the control throughout the designed range of control movement.

5.6.6 Limit speed. Do not exceed applicable speed for particular maneuvers (e.g., V_H, V_{NE}, V_{DL}).

5.6.7 DNE limits on any critical component. During any specific maneuver, it is possible to achieve DNE limits on a component without attaining any other cutoff limit. At a minimum, return the aircraft to the control area for evaluation prior to resuming flight if DNE limits are exceeded. Depending on exceedance, aircraft may be directed to land as soon as possible. Exceedance of a DNE limit requires investigative action, a damage evaluation, and special disposition of all affected components.

5.6.8 Travel limits of dynamic components. Rotor flapping or teetering limits attaining 95 percent of their static full-throw values. If stop contact is made, the aircraft should be returned to the control area for inspection prior to resuming flight.

5.6.9 Difficult or unsafe recover maneuver. Any maneuver the aircrew judges would result in an unusually difficult or unsafe recovery.

5.6.10 Exceedance of red-line. No established red-lines should be exceeded except as specified herein, by the contractor flight release, or in an emergency.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This standard should be used to verify the requirements of MIL-STD-3063. Though requirements are contained outside this document, the following scenarios are a suggested subset of events that may drive the need for a structural demonstration:

a. New aircraft design.

b. Performance related modification to existing aircraft that alters the performance envelope (V-N diagram). One such example is an engine upgrade that increases power available.

c. Structural modification to existing aircraft that alters the structural envelope (V-N diagram). One such example is a rotor head redesigned to carry greater loads.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this standard.

6.3 Tailoring guidance for contractual application. To ensure proper application of this test method standard, tailored applications of the requirements in sections 4 or 5 of this standard should require concurrence of the procuring Government agency, PEO, service designated user representatives, and the service design control activity (MAA). Tailored applications of this standard to joint rotorcraft programs should require concurrence of user representatives and design control activities for each service. For existing contracts, further tailoring of this test method standard should also require concurrence of contractor designated technical fellows or safety committee representing a structural integrity standpoint.

6.4 Supersession data. This standard supersedes AMACC Appendix K, *Airworthiness Qualification Requirements, Structural Demonstration for Rotorcraft*, and ADS-24. Any reference to either document should instead use this standard.

6.5 Documentation. As a minimum, the structural demonstration report should include time histories, a discussion of pilot technique used in each maneuver, a summary table showing the maximum measured load for each structural component along with flight condition and maneuver of occurrence, a complete table of maneuvers with applicable flight data, and the flight log. Selected build ups, which represent varying techniques, should also be included. All provided documentation should be IAW the method of compliance of Structural Demonstration section of AMACC. Interpret and evaluate test results IAW applicable section of MIL-STD-3063. Determine and document any operating restrictions required to prevent loads from exceeding those used in structural analyses and static tests.

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6.6 Test data delivery. Recorded on-board test data should be retained and delivered both in original "raw" format with necessary equations or algorithms for derived parameters and post-processing and in readable post-processed format agreed upon by the procuring Government agency. At present, IRIG 106-22 is the preferred applicable telemetry data standard, with Chapter 10 as the digital recording standard. Recommended post-processed data file should include aircraft state data, control positions, bus parameters, and strain gage data.

6.7 Disposition of flight test rotorcraft. The damage fraction for all safety of flight structure, principal structural elements (PSEs), or Critical Safety Items (CSIs)[9] should be established for the test aircraft. Any DNE exceedances and associated disposition should be documented. Detailed inspection of the aircraft for fatigue cracking, excessive wear, and damage should be performed. The OEM should make a recommendation with supporting justification for the disposition of all safety of flight structure, PSEs, or CSIs (e.g., acceptable for service use as-is, acceptable for service use with flight hour adjustment added to component service life, retain only for test use, scrap part, or other disposition). The Government will establish test article disposition after review of recommendations and rationale provided by the OEM.

6.8 Test plan guidance. The flight envelope should encompass the complete range of parameters to be allowed for operational usage to include altitude, airspeed, load factor, gross weight, CG, configuration, etc. Sufficient checkout and buildup flying should be conducted, using the control techniques peculiar to the demonstration, to give reasonable assurance of successful test completion. Specific control inputs should be defined in a maneuver standards guide or test plan and are necessary to gain consistency in the results by eliminating, to every practical degree, differences in pilot technique and expertise.

6.9 Basis for SDGW. The SDGW should be the highest flight weight, with full internal and external load items (including full fuel) required for it to perform the primary mission defined in the aircraft performance specification, required for the maximum positive and minimum negative load factors of maneuvering flight.

6.10 Basis for MAGW. The MAGW represents the aircraft configuration with the maximum weight and mass distribution without a flight restriction for maneuver severity. Weights above this maximum are limited by rotorcraft strength requirements.

6.11 Historical rotorcraft maneuvers. In ADS-24 that serves as a basis for this standard, precise numerical values were agreed upon at that time by a multidisciplinary group, primarily applicable to a single specific rotorcraft platform. Future values should be agreed upon by the OEM and procuring Government agency and documented in the test plan, particularly with input from persons with operational, performance and handling qualities, and structural backgrounds.

6.12 Other sources for maneuver details. ADS-29 and ADS-33-PRF also contain maneuver details; however, those documents have different objectives (structural design versus handling qualities, respectively) when compared to this standard. As such, direct use of ADS-29 and ADS-33-PRF as

[9] See MIL-STD-3063, Categories of structural elements. Depending on if aircraft is still in design vs sustainment phase, CSIs may not yet be defined.

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maneuver guides without consideration for structural demonstration objectives is not recommended.

6.13 References. This section includes documents recommended for additional information or as examples. See section 2 for documents required in sections 3, 4, and 5.

a. ADS-24 (inactive), *Structural Demonstration*, 1 November 1985. (Copies of canceled or inactive U.S. Army Aeronautical Design Standards may be made available by contacting POCs at https://www.avmc.army.mil/Directorates/SRD/TechDataMgmt/.)

b. ADS-29 (inactive), *Structural Design Criteria for Rotary Wing Aircraft*, September 1986. (Copies of canceled or inactive U.S. Army Aeronautical Design Standards may be made available by contacting POCs at https://www.avmc.army.mil/Directorates/SRD/TechDataMgmt/.)

c. ADS-33-PRF, *Handling Qualities Requirements for Military Rotorcraft*, 21 March 2000. (Copies of this document are available online at https://www.avmc.army.mil/Directorates/SRD/TechDataMgmt/.)

d. AMCP 706-203, *Engineering Design Handbook, Helicopter Engineering, Part Three, Qualification Assurance*, April 1972. (Copies of this document are available online at https://apps.dtic.mil/docs/citations/AD0901657.)

e. MIL-S-8698 (canceled), *Structural Design Requirements, Helicopters*, 1 July 1954. (Copies of this document are available online at https://quicksearch.dla.mil).

f. IRIG Standard 106-22, *Telemetry Standards*, May 2022. (Copies of this document are available online at http://www.irig106.org/.)

6.14 Subject term (keyword) listing.

Airspeed, maximum
Arrestment, hover turn
Envelope, structural design (V-N diagram)
Landing, hard
Load factor, maximum
Overspeed, autorotation
Pullout, autorotative
Pullout, rolling
Pullout, symmetrical
Pushover, symmetrical
Restrictions, operating
Reversals, turn
Sideslip, steady
Stop, quick

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